

DO NOT WRITE ON TEST



**FINANCIAL MATH &
ANALYSIS CONCEPTS**

~OPEN EVENT~

(190)

STATE 2024

TOTAL POINTS

_____ 100 Points

Test Time: 60 minutes

GENERAL GUIDELINES:

Failure to adhere to any of the following rules will result in disqualification:

1. Contestant must hand in this test booklet and all printouts if any. Failure to do so will result in disqualification.
2. No equipment, supplies, or materials other than those specified for this event are allowed in the testing area. No previous BPA tests and/or sample tests (handwritten, photocopied, or keyed) are allowed in the testing area.
3. Electronic devices will be monitored according to ACT standards.
4. For any interest calculations, assume a 360-day year.
5. Unless otherwise stated, assume that all plant assets were purchased on January 1.
6. Unless otherwise stated, assume that the overtime rate is 1.5 regular time pay for any hours worked in excess of 40 hours per week.

Multiple Choice Questions

Directions: Identify the letter of the choice that best completes the statement or answers the question.

1. Tom and Julie budgeted \$7,500 to remodel their living room. They spent \$8,201. Percentage wise, how much did they go under or over budget?
 - A. 9.3% Under Budget
 - B. 9.4% Over Budget
 - C. 9.3% Over Budget
 - D. 9.4% Under Budget
2. A forklift that costs \$15,400 is estimated to depreciate 12% per year. What is the book value of the forklift at the end of the first year?
 - A. \$1,848
 - B. \$13,552
 - C. \$15,400
 - D. \$14,117
3. Jolene Timmer started her cash drawer with \$80 at the beginning of the day. The cash register reading at the end of the day showed that she had \$3,820.10 in credit card sales, \$1,104.57 in cash sales. If the cash on hand in the drawer is \$1,183.17, does that match the amount that it should be?
 - A. No, there is a cash over of \$1.40
 - B. No, there is a cash under of \$1.40
 - C. Yes, the cash matches perfectly
 - D. There isn't enough information to decide
4. Robert's employer provides health insurance. The monthly premium is \$1,740 and Robert is responsible for 20% of the cost. How much does Robert's employer pay each month?
 - A. \$2,088
 - B. \$1,740
 - C. \$348
 - D. \$1,392
5. Vincent Costello has two job offers and isn't sure which one to accept. Job One pays an hourly rate of \$18.50 per hour and involves working 40 hours per week. Job Two pays \$15.25 and requires 8 hours per day for 5 days each week. Another employee working the same job says she earns about \$50 per day in tips. Based on this information, which job should Vincent accept and why?
 - A. Job 1, because he'll earn \$120 more each week
 - B. Job 2, because he'll earn \$120 more each week
 - C. Job 1, because he'll earn \$130 more each week
 - D. Job 2, because he'll earn \$130 more each week

6. A pair of shoes with a list price of \$46 is sold to a customer for \$29.90. What was the rate of discount given to the customer?
 - A. 35%
 - B. 54%
 - C. 62%
 - D. 65%
7. Everything Crafts marks their merchandise up 40% above cost. If the cost price of a basket at the store is \$16.70, what would Everything Crafts charge for it?
 - A. \$6.68
 - B. \$16.70
 - C. \$23.38
 - D. \$17.10
8. Allen Olsen earns \$16.80 an hour and his employer gives him a Cost-of-Living Adjustment (COLA) of .8%. What will Allen earn after the COLA?
 - A. \$16.83
 - B. \$20.16
 - C. \$16.93
 - D. \$30.24
9. The Michaels family is planning to purchase a new home for \$284,000. If they make a down payment of 20 percent, what will be the amount of their loan principle?
 - A. \$227,200
 - B. \$340,000
 - C. \$56,800
 - D. \$284,000
10. The management of a company plans to share 20% of its \$717,400 profit with its 68 employees. How much will each employee receive?
 - A. \$6,660
 - B. \$8,440
 - C. \$10,550
 - D. \$2,110
11. Leo Cornwall could buy a digital camera for \$1,500 or buy it for \$100 down and the balance in 24 monthly payments of \$70.57. How much greater is the installment price than paying cash?
 - A. \$193.68
 - B. \$293.68
 - C. \$1,693.68
 - D. \$1,793.68

12. Brandon's net pay is \$5,000 per month. Brandon's rent on his apartment is \$1,200, his car payment is \$350, and his car insurance is \$120. Brandon spends about \$150 a month on gas, \$200 on groceries, and \$400 on dining out. How much of Brandon's net pay is spent on fixed expenses?
- A. 48.4%
 - B. 15.0%
 - C. 33.4%
 - D. 38.0%
13. An office has an area of 2,500 sq. ft. and costs \$62,250 in rent each year. Utilities cost \$4,800 per year and maintenance is \$1,750 per year. What is the yearly cost of this office space per square foot?
- A. \$27.52
 - B. \$24.90
 - C. \$1.92
 - D. \$0.70
14. Joslyn Sullivan works for Wydner Windows and gets paid bi-weekly. For the most recent pay period, she earned \$602.27 gross pay. Joslyn's deductions include Federal Income Tax of \$13.00, State Tax of 4.25%, Social Security of 6.2%, Medicare of 1.45%, and a deduction for health insurance. For health insurance, Joslyn pays 20% of her annual insurance premium of \$7,618. What is Joslyn's net pay?
- A. \$517.60
 - B. \$380.40
 - C. \$467.12
 - D. \$459.00
15. Claire Andrews put 1,380 miles on her car in August. Of those miles, 18% were for her job. If her employer reimburses Claire \$0.48 per mile for business use, how much will she get reimbursed for use of her car in August?
- A. 662.40
 - B. 344.45
 - C. 119.23
 - D. 543.17
16. Which of the following is considered the best credit score?
- A. 320
 - B. 270
 - C. 650
 - D. 780
17. A person that gets paid bi-monthly gets paid _____ times per year.
- A. 6
 - B. 24
 - C. 26
 - D. 12

18. Joan's taxable income last year was \$34,800. The graduated income tax table below shows the tax schedule for Joan's state of residence. How much did she pay for state income taxes?

For Taxable Income		
Over--	But Not Over--	The Tax is--
\$0	\$8,000	2% of taxable income
\$8,000	\$16,000	\$160 plus 3% of taxable income over \$8,000
\$16,000	\$24,000	\$400 plus 4% of taxable income over \$16,000
\$24,000	\$32,000	\$720 plus 5% of taxable income over \$24,000
\$32,000	\$40,000	\$1,120 plus 6% of taxable income over \$32,000
\$40,000	\$48,000	\$1,600 plus 7% of taxable income over \$40,000
\$48,000	-----	\$2,160 plus 8% of taxable income over \$48,000

- A. \$2,088
B. \$1,120
C. \$1,288
D. \$952
19. Billy is buying a 2,000 square foot house for \$225,000. What is the price per square foot?
- A. \$200.00
B. \$225.00
C. \$112.50
D. \$88.89
20. A 20 oz soda costs \$2.25 and a 30 oz soda costs \$2.40. What is the cost savings per ounce of the 30 oz soda?
- A. \$0.15
B. \$0.0325
C. \$0.08
D. \$0.11
21. Marty works a 40-hour week earning \$18.95 an hour. He also receives fringe benefits valued at 19% of his gross wages and has the following yearly job expenses: Commuting, \$456.00; Union Dues, \$750; and Parking, \$470.00. What is the yearly net value of Marty's job?
- A. \$45,229.04
B. \$37,740.00
C. \$48,581.04
D. \$39,416.00
22. Daisy purchased a new computer for \$225. The computer has a 10% discount, and sales tax is 8%. How much will Daisy pay in total for the computer?
- A. \$243.00
B. \$229.50
C. \$218.70
D. \$227.70

23. If Mark gets paid \$12.43 per hour, what is his overtime rate if he gets paid double overtime?
- A. \$18.645
 - B. \$12.43
 - C. \$18.65
 - D. \$24.86
24. Carter Blanchard borrowed \$500 on a one-year simple interest installment loan with 18% interest. His monthly payments are \$45.84. How much of his first payment is applied to the principal?
- A. \$7.50
 - B. \$38.34
 - C. \$45.84
 - D. \$90.00
25. Dynamic Industries manufactured 150 phones. The related manufacturing costs were \$1,282.29 for materials, \$1,975.26 for labor, and \$1,234.31 for factory overhead. What was the total cost to manufacture each phone to the nearest cent?
- A. \$21.70
 - B. \$21.72
 - C. \$29.95
 - D. \$29.90
26. Devon Thurman went on a 3-day business trip and had the following expenses: airfare, \$586; shuttle service, \$48; meals, \$217; hotel, \$189 a night for three nights, and \$300 in conference fees. What was the amount of his reimbursement?
- A. \$1,340
 - B. \$1,214
 - C. \$1,529
 - D. \$1,718
27. Denson, Inc. pays \$30,000 each month to rent its factory. The cost of rent is distributed among its departments based on the floor space used by each. The Sales & Marketing Department used 1,500 sq. ft., the Administrative Department uses 3,000 sq. ft., The Finance Department uses 3,500 sq. ft., and the Technology Department uses 4,000 sq. ft. of floor space. How much should be charged to the Technology Department?
- A. \$7,500.
 - B. \$3,750
 - C. \$10,000
 - D. \$8,750

28. Rhonda earns \$3,750 per month. From these wages, \$312 in federal withholding, \$245.25 in Social Security, \$59 in Medicare, \$104 in state income taxes, and \$23.75 in health insurance are deducted. Find the percentage of gross pay that Rhonda takes home to the nearest whole percent.
- A. 65%
 - B. 80%
 - C. 70%
 - D. 85%
29. Donna Valentine earns \$43,877 working as an administrative assistant for Shepard Law Firm. This month she will be getting a cost-of-livelihood raise of 2.5%. In addition, her employer is giving her a 4% merit raise because of her exceptional job performance. What will Donna's new salary be for the year (rounded to the nearest dollar)?
- A. \$46,729
 - B. \$44,974
 - C. \$45,632
 - D. \$46,773
30. Glitter Boutique sold 5 greeting cards at \$3.75 and two bracelets at \$12.99 each on Invoice #2816. What was the total amount of the sales invoice?
- A. \$16.74
 - B. \$24.24
 - C. \$44.73
 - D. \$72.45
31. Slick Disc sold a CD for \$8.59. The store's cost of the DVD was \$4.15. What was the percentage markup based on the selling price?
- A. 106.99%
 - B. 48.3%
 - C. 93.5%
 - D. 57.1%
32. A \$300,000 term life insurance policy is sold at a rate of \$1.18 per \$1,000 of insurance. What is the annual premium?
- A. \$1,180
 - B. \$354
 - C. \$236
 - D. \$177
33. Shelly Winston made the following deposit: 72 twenties; 126 tens; 57 fives; 235 ones; 287 quarters; 312 dimes; 48 nickels; 347 pennies; and checks for \$124.68, \$132.08, \$1.29, and \$5.79. She asked for a fifty-dollar bill back in cash. What was her total deposit?
- A. \$3,642.66
 - B. \$3,542.66
 - C. \$3,592.66
 - D. \$3,278.82

34. Kim Beale earned \$24,800 in taxable income this year. She pays state taxes according to the following tax schedule. How much did Kim pay for state taxes this year?

State Tax Rates
1% of the first \$5,000 of taxable income
2% of the next \$15,000 of taxable income
3% of the next \$20,000 of taxable income
4% of all taxable income over \$40,000

- A. \$744
B. \$496
C. \$400
D. \$494
35. Blake earns \$4,500 per month and has expenses of \$3,000. If Blake sets aside 10% of his net earnings each month, how long will it take him to build up an emergency savings fund that will cover 6 months of his expenses?
- A. 2 months
B. 20 months
C. 40 months
D. 60 months
36. Camille's rent is \$450 a month, and her renter's insurance is 4.5% of her monthly rent. How much is Camille's annual renter's insurance premium?
- A. \$54.00
B. \$2,025
C. \$20.25
D. \$243.00
37. Andrew Martin placed \$1,500 in a six-month certificate of deposit (CD) earning quarterly interest an annual rate of 4%. How much interest will Andrew receive each quarter?
- A. \$6
B. \$15
C. \$30
D. \$60
38. Melanie Stirling works for straight commission and sells two types of fountains. The small fountain sells for \$869, and she earns a commission of \$174. The large fountain sells for \$1,260, and she earns \$252 in commission. Over the last 3 months Melanie sold 19 small fountains and 23 large fountains. What did Melanie earn in commission during the last 3 months?
- A. \$45,491
B. \$8,790
C. \$9,102
D. \$36,389

39. The table below shows freight weights for Express Shipping, Inc. What is the freight charge for a 1,500-pound package to be delivered in 3 days to a business in zone 5?

Express Shipping, Inc.							
3-Day Delivery—Rates Per Pound							
Weight in Pounds	Destination Zones						
	1 & 2	3	4	5	6	7	8
151-499	0.50	0.90	1.10	1.19	1.66	2.08	2.34
500-999	0.49	0.88	1.07	1.17	1.61	1.97	2.34
1,000-1,999	0.47	0.85	1.04	1.14	1.56	1.94	2.28
2,000 +	0.45	0.83	1.03	1.09	1.51	1.92	2.23

- A. \$1,560
B. \$2,340
C. \$1,755
D. \$1,710
40. Betty Thomas owns a craft store and plans to sell 260 Welcome Signs this year. She earns \$25 for every Welcome Sign sold. During the first 8 months of the year Betty sold an average of 16 Welcome Signs and earned \$3,200. How many Welcome Signs must she sell on average in each of the remaining 4 months to reach her goal?
- A. 33
B. 44
C. 128
D. 132
41. What is the median of the following set of numbers: 1,15,7,3,12,8,5?
- A. 5
B. 7
C. 8
D. 10
42. Joe Valentine just started a new sales job that pays 4% straight commission on all sales. On average, sales in his territory are \$20,400 per week. Assuming sales continue at this rate, how much could Joe earn after working one year?
- A. \$816
B. \$42,432
C. \$42,000
D. \$43,000

43. Kohler offers a faucet with a list price of \$140 to retailers with a series discount of 20%, 20%, and 5%. What is the invoice price of one faucet?
- A. \$77
 - B. \$112
 - C. \$54.88
 - D. \$85.12
44. Gary Lawson purchases a new car with a purchase price of \$23,340. He is charged 6% sales tax on the purchase price and will get a manufacturer's rebate of \$1,250. He also must pay \$128 for registration. What is the balance due on the car?
- A. \$24,876.08
 - B. \$23,618.40
 - C. \$23,354.72
 - D. \$26,118.40
45. A waitress at an exclusive restaurant presented the customer with a food and beverage check for \$340. The customer left a 20% tip. How much did the customer pay in total?
- A. \$346.80
 - B. \$68.00
 - C. \$408.00
 - D. \$374.00
46. Angelica Bardo invested \$5,500 into a 3-year certificate of deposit (CD) earning an annual 4% simple interest rate. She cashed in the CD before the end date with an early withdrawal penalty of six month's interest. What was the value of Angelica's investment after the penalty was deducted?
- A. \$6,160
 - B. \$6,050
 - C. \$5,390
 - D. \$6,270
47. The Munns Company has 10 employees. Five of the employees earn \$12 an hour, one earns \$11 an hour, two earn \$9 an hour, and two earn \$10 an hour. What's the average amount per hour the employees are paid?
- A. \$11.00
 - B. \$9.80
 - C. \$10.00
 - D. \$10.90
48. Paula Jones worked the following hours over five days: $8\frac{1}{4}$, $7\frac{1}{2}$, $10\frac{1}{4}$, $8\frac{3}{4}$, and $9\frac{1}{2}$. If she earns \$15.60 per hour and overtime is based on an 8-hour day, how much did Paula receive in overtime pay?
- A. \$111.15
 - B. \$66.30
 - C. \$74.10
 - D. \$99.45

49. Deanna Milford had total sales of \$180,700 last month and earned a commission of \$5,040. She earns commission on monthly sales over \$50,000. What is her rate of commission rounded to the nearest tenth percent?
- A. 2.7%
 - B. 2.8%
 - C. 3.8%
 - D. 3.9%
50. Dee Glover earns \$500 each week and has the following amounts deducted: \$38 for federal taxes, \$38.25 for FICA taxes, \$30 for state taxes, \$84 for health insurance, and \$40 for union dues. To the nearest percent, what percentage of her gross pay does Dee take home?
- A. 50%
 - B. 54%
 - C. 46%
 - D. 78%